

Analyzing and Improving Tip-of-the-Tongue Retrieval with Neural Re-Ranking: Final Report

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Abstract—Tip-of-the-Tongue (ToT) retrieval is a known-item retrieval task where users describe a target through incomplete and indirect memories rather than exact titles or keywords. This project investigates neural re-ranking on the TREC-ToT 2023 development set using BM25, dense retrieval, dense-lexical hybrid retrieval, Reciprocal Rank Fusion, CrossEncoder re-ranking, rank-preserving fusion, and multiview scoring. The results show that candidate generation is a major bottleneck: expanding BM25 from top-100 to top-1000 substantially improves candidate recall, and hybrid retrieval recovers additional relevant documents. However, these gains do not directly yield strong top-rank improvements. Pure CrossEncoder re-ranking improves Recall@100, reaching 0.2133 over BM25@1000, but reduces MRR@10 and nDCG@10. Rank-preserving fusion gives the best top-rank results, though gains over BM25 remain small and statistically non-significant. Contrastive rank-movement analysis shows that many failures come from candidate absence, while the CrossEncoder often over-promotes semantically plausible distractors. Overall, ToT retrieval requires stronger candidate generation and task-specific re-ranker calibration rather than direct use of off-the-shelf neural re-ranking.

I. INTRODUCTION

Tip-of-the-tongue (ToT) retrieval [1] is a known-item retrieval task in which the user attempts to find a specific target without exactly remembering its title, name or canonical keywords. Instead of issuing a precise keyword query, the user describes it through their partial memory, vague contextual clues, uncertain plot details, or approximate semantic associations. This makes the task fundamentally different from standard retrieval. In standard ad-hoc retrieval, many documents may satisfy the information need, whereas in ToT retrieval, the system must recover a specific known item from an incomplete and noisy description.

This setting is challenging for both lexical and neural retrieval models. Lexical models such as BM25 [2] are effective first-stage retrievers, but they depend on term overlap. In ToT retrieval, users may describe a film, book, or character without using vocabulary that appears in the target document, causing the relevant item to be ranked too deeply or excluded from shallow candidate pools. Neural models can capture broader semantic similarity, but they may also drift toward topically plausible distractors rather than the specific known item intended by the query.

The initial goal was to improve ToT retrieval with a two-stage neural re-ranking pipeline: BM25 retrieves a candidate

pool, and a CrossEncoder reorders it using query-document interaction modeling. This combines BM25’s efficiency and lexical precision with the semantic matching ability of transformer re-rankers. However, it depends on two conditions: the relevant document must first appear in the candidate pool, and the re-ranker must promote it without disrupting useful lexical rankings.

The final phase of this project therefore investigates the retrieval pipeline diagnostically rather than treating re-ranking as a black box improvement over BM25. The central question is not only whether a CrossEncoder improves BM25, but why it succeeds or fails under different candidate generation and fusion strategies. To answer this, I evaluate BM25 candidate expansion, dense retrieval, dense-lexical hybrid candidate generation, RRF, pure CrossEncoder re-ranking, rank-preserving neural fusion, alternative re-rankers, confidence-gated re-ranking, field-aware document representations, and multiview CrossEncoder scoring.

The experiments are organized around four research questions inspired by the first phase of the project.

- Does transformer-based neural re-ranking improve retrieval effectiveness over a BM25 baseline for Tip-of-the-Tongue queries?
- To what extent does first-stage candidate availability limit neural re-ranking in ToT retrieval?
- Can dense retrieval complement BM25 by recovering relevant documents that are missing from the BM25 candidate pool?
- Can controlled re-ranking strategies such as rank-preserving fusion, confidence gating, and multiview scoring reduce the instability of pure neural re-ranking?

The results show a separation between candidate generation and re-ranking. Larger BM25 pools improve candidate recall, meaning BM25 often ranks the target too deeply rather than missing it entirely. Dense retrieval is weak alone but adds complementary candidates. However, better candidate availability does not directly improve top-rank metrics. Pure CrossEncoder re-ranking raises Recall@100 in some settings, but lowers MRR@10 and nDCG@10 by promoting plausible distractors. Rank-preserving fusion stabilizes the neural model with a strong lexical prior, giving the best top-rank results, though the gains are small and statistically non-significant.

Overall, this project shows that ToT requires more than simply adding an off-the-shelf neural re-ranker to BM25. Candidate expansion creates more opportunities for retrieval, but generic neural re-rankers are not calibrated for this type of a task. The main contribution of this study is therefore a systematic analysis of where the BM25-plus-neural-reranking pipeline fails: at candidate generation, at semantic re-ranking, and at the interaction between lexical and neural signals.

II. DATASET

The experiments are conducted on the Tip-of-the-Tongue 2023 development set. The collection contains 150 queries, 231k documents, and one known relevant document for each query. This makes the task a known item retrieval problem. For each query, the goal is not to retrieve a wide set of relevant documents, but to recover the specific item that the user is trying to remember.

Each query contains an indirect description of the target item. These descriptions create a mismatch between the query text and the document text, since the query often does not contain the exact title or keywords of the target document.

The document collection consists of structured JSONL records. Each document includes an identifier, and textual fields such as the page title and main text. Some documents also contain additional information such as sections and infobox-style metadata. For the experiments, documents are represented as concatenated text. This representation is intended to approximate the full textual evidence available for the retriever. For field-aware and multiview experiments, structured components such as title, synopsis, reception sections are considered separately to test whether neural re-ranking benefits from more focused document views.

III. EVALUATION SETUP

The task is evaluated using ranking metrics that reflect both early precision and candidate availability. Since each query has one known relevant document, the rank position of that document is central to the evaluation.

The primary metrics are $MRR@10$, $nDCG@10$, and $Recall@100$. MRR measures whether the target document appears in the top-10 results. This is especially important because users are unlikely to inspect many results when trying to identify a known item. $nDCG$ also evaluates the quality of the top-10 ranking, applying a stronger reward when the relevant document appears closer to rank 1. Lastly, $Recall$ measures whether the target document appears anywhere in the first 100 retrieved documents.

In addition to these metrics, I also computed Candidate Recall@k for first stage retrieval. Candidate Recall measures whether the relevant document appears within the candidate pool before neural re-ranking. This is important because a re-ranker cannot recover a document that is not present in its input candidate set, indicating a specific bottleneck. Candidate Recall@K is evaluated at $K = 100, 500$, and 1000 for BM25, and at comparable depths for dense and hybrid candidate generation.

For statistical testing, I use paired query-level comparisons. $MRR@10$ and $nDCG@10$ are compared with the Wilcoxon signed-rank test because they produce continuous per-query scores and the same query set is evaluated under each system. Candidate recall comparisons are evaluated with an exact McNemar test because candidate availability is a paired binary outcome: for each query, the relevant document is either present or absent from a candidate pool. This separates improvements in candidate generation from improvements in final ranking quality.

The evaluation therefore distinguishes three related but different questions: whether the system retrieves the target document at all, whether the target document appears in a candidate pool available to the re-ranker, and whether the re-ranker can promote the target document into the top ranks.

IV. MODELING AND METHODOLOGY

The retrieval pipeline is organized into two stages: First-stage candidate generation and second-stage neural re-ranking. The first stage determines which documents are made available to the re-ranker, while the second stage determines whether those candidates can be reordered effectively. This separation is important because a poor final ranking may result either from candidate absence or re-ranking failure.

A. BM25 Baseline

The baseline model is BM25, implemented with rank-bm25. BM25 is used as the main lexical first stage retriever because it is efficient, interpretable and competitive for sparse tasks. A parameter sweep is conducted over different values of k_1 and b , and the best configuration is selected according to validation performance on ToT 2023 development set. The best performing configuration was $k_1=1.6$ and $b=0.9$.

The BM25 baseline is evaluated at a retrieval depth of 1000, while final metrics are computed at $MRR@10$, $nDCG@10$ and $Recall@100$. This allows the same BM25 ranking to be used both as a final retrieval baseline and as an input candidate pool for re-ranking experiments.

B. Candidate Pool Expansion

The initial re-ranking setup used the top-100 BM25 candidates as input at first. However, this creates a highly strict upper bound. If the relevant document is not in the top-100, re-ranking cannot recover it. Therefore, I evaluate BM25 candidate recall at multiple depths: top-100, top-500 and top-1000.

This experiment is not intended as a new ranking model by itself. Instead, it diagnoses whether the primary limitation is candidate availability. If the relevant document appears at deeper BM25 ranks, then the problem is not complete lexical failure but insufficient candidate depth and re-ranker calibration.

C. Dense Retrieval

To test whether semantic retrieval can recover documents that BM25 misses, I implement a dense retrieval baseline using

a SentenceTransformer [3] model. Documents and queries are set to encode independently and candidate documents are retrieved using vector similarity. Dense retrieval is evaluated both as a standalone retriever and as a source of additional candidates.

The purpose was not only to compare dense retriever to BM25, but also to measure complementarity. A dense retriever may perform poorly as a standalone re-ranker while still retrieving relevant documents that are missing from the BM25 pool. Such cases are crucial because they indicate that lexical and semantic signals fail differently.

D. Dense-Lexical Hybrid Candidate Generation

I evaluate hybrid candidate generation by combining two retrieval outputs. The main strategy is candidate union, combining BM25@100 with Dense@1000, preserving the BM25 candidates and appending dense candidates that are not already in BM25 candidate pool. This design protects strong lexical ranking while increasing candidate availability.

I also evaluate RRF (Reciprocal Rank Fusion) [5] as an alternative hybrid strategy. Unlike simple union, RRF directly combines the ranks from BM25 and dense retrieval. This tests whether dense ranking should influence the final ordering directly or whether they are more useful as a candidate expansion mechanism.

E. Pure CrossEncoder Re-ranking

The main neural re-ranking model is a CrossEncoder trained for passage ranking. Unlike dense retrieval, a CrossEncoder receives the query and document together and produces a relevance score from their joint representation. This makes it a natural second stage model for re-ranking a limited candidate pool.

Pure CrossEncoder re-ranking fully replaces the original BM25 or hybrid ranking within the candidate pool. Candidates are scored independently from prior scores and sorted according to neural relevance scores. This experiment tests whether the neural model can act as a complete re-ranker after candidate generation.

F. Rank-Preserving Neural Fusion

Since pure CrossEncoder re-ranking can disrupt useful lexical rankings, I also evaluate rank-preserving fusion. In this approach, the final score combines the original rank-based prior with the normalized CrossEncoder score

$$s_{\text{final}} = \lambda \cdot s_{\text{BM25}} + (1 - \lambda) \cdot s_{\text{CE}} \quad (1)$$

Here, rank score is derived from the original candidate rank, while CE score is the normalized neural re-ranking score. Higher values of λ preserve the lexical or hybrid ranking more strongly. I evaluate several λ values, including 0.90, 0.95, 0.98, and 0.99. This tests whether the neural re-ranker is more effective as a small corrective signal rather than as a full rank replacement.

G. Controlled Re-Ranking Variants

In addition to the main CrossEncoder and fusion experiments, I conduct several targeted ablations to understand why neural re-ranking succeeds or fails.

First, I test an alternative neural re-ranker, BGE, to determine whether the observed limitations are specific to the MiniLM [4] CrossEncoder or reflect a broader mismatch between off-the-shelf re-rankers and ToT retrieval.

Second, I test rank injection, where BM25 rank information is inserted into the CrossEncoder input text. This experiment investigates whether explicitly exposing lexical rank information helps the neural model learn to respect the original retrieved signal.

Third, I test confidence-gated re-ranking. Instead of allowing the CrossEncoder to reorder all candidates freely, this method promotes candidates only when the neural score exceeds a confidence threshold. The goal is to reduce destructive re-ranking caused by uncertain neural scores.

Fourth, I test field-aware re-ranking, where documents are represented using structured fields such as title, abstract-like sections, synopsis, and infobox metadata instead of the full concatenated text. This tests whether a more compact and focused document representation improves neural matching.

Finally, I evaluate multiview CrossEncoder scoring. Rather than choosing one document representation, each document is scored through multiple views, such as full text, title and abstract, synopsis, metadata, compact representation, and reception. The final neural score is obtained using view aggregation, such as maximum score or a combination of maximum and mean score. This tests whether the relevant ToT clue may appear in only one part of the document and can be better captured by view-specific scoring.

V. RESULTS

The experiments show a clear distinction between candidate generation gains and final top-rank effectiveness. While expanding and diversifying the candidate pool substantially increases the availability of relevant documents, neural re-ranking does not consistently convert this additional availability into higher MRR@10 or nDCG@10. The results are therefore reported in three groups: first-stage candidate generation, CrossEncoder-based re-ranking, and controlled re-ranking variants.

A. BM25 Baseline and Candidate Availability

The tuned BM25 baseline uses $k1 = 1.6$ and $b = 0.9$. This configuration achieves $\text{MRR@10} = 0.0710$, $\text{nDCG@10} = 0.0747$, and $\text{Recall@100} = 0.1533$. Although this baseline is weak in absolute terms, it provides the strongest sparse lexical reference point for the remaining experiments.

A candidate pool expansion analysis shown in Table I demonstrates that shallow BM25 retrieval is a major bottleneck. In the table, Cand. Rec. denotes candidate recall, Δ is the change relative to BM25@100, and New indicates newly recovered queries. BM25@100 retrieves the relevant documents for only 15.33% of queries. Increasing the pool to BM25@500

TABLE I
CANDIDATE AVAILABILITY ANALYSIS.

Pool	Cand. Rec.	Δ	New	p
BM25@100	0.1533	—	—	—
BM25@500	0.2267	+0.0733	11	0.000977
BM25@1000	0.3000	+0.1467	22	< 0.001
Dense@1000	0.1733	+0.0200	17	0.7201
Hybrid@1000	0.2667	+0.1133	17	0.000015

raises candidate recall to 22.67% and BM25@1000 raises it to 30%. The improvement from top-100 to top-500 is statistically significant according to McNemar’s exact test ($p=0.000977$), and the improvement from top-100 to top-1000 is also significant ($p < 0.001$). This indicates that BM25 often does not fail completely; instead, it ranks the relevant document rather too deeply for a shallow re-ranking setup.

An oracle upper-bound analysis further confirms this bottleneck. If a perfect re-ranker moved the relevant document to rank 1 whenever it appeared in the candidate pool, the oracle MRR@10 would be 0.1533 for BM25@100, 0.2267 for BM25@500, and 0.3000 for BM25@1000. The actual BM25 MRR@10 remains 0.0710, showing that a larger candidate pool creates opportunity, but the re-ranker must still identify the correct document among many distractors.

B. Dense and Hybrid Candidate Generation

Dense retrieval performs poorly as a standalone ranker. The dense system obtains MRR@10 = 0.0131, nDCG@10 = 0.0178, and Recall@100 = 0.0667. This is substantially below the BM25 baseline. However, dense retrieval is still useful as a complementary candidate source. Dense@1000 recovers relevant documents for 17 queries that are missed by BM25@100, although it also misses 14 queries that BM25@100 retrieves.

The BM25-dense hybrid union therefore improves candidate availability without sacrificing BM25’s top-ranked candidates. The BM25@100 and Dense@1000 hybrid union preserves the BM25 baseline scores at the top of the ranking, with MRR@10 = 0.0710, nDCG@10 = 0.0747, and Recall@100 = 0.1533, while increasing candidate recall at depth 1000 to 0.2667. This improvement over BM25@100 is statistically significant ($p = 0.000015$), and it adds 17 newly recovered queries without losing any of the BM25@100 successes.

In contrast, Reciprocal Rank Fusion is less effective for final ranking. Hybrid RRF obtains MRR@10 = 0.0595, nDCG@10 = 0.0706, and Recall@100 = 0.1533. Although RRF preserves the broader candidate availability of the hybrid pool, allowing dense ranks to directly influence the ordering disrupts the stronger BM25 top-rank signal.

C. CrossEncoder Re-Ranking

Pure CrossEncoder re-ranking does not consistently improve top-rank effectiveness. When applied to the BM25@1000 candidate pool, pure CrossEncoder re-ranking obtains MRR@10 = 0.0256 and nDCG@10 = 0.0366, which are far below the BM25 baseline. However, Recall@100 increases to 0.2133, the highest Recall@100 observed among the evaluated systems.

This shows that the CrossEncoder can identify and promote some relevant documents into the top 100, but it does not rank them accurately enough for top-10 effectiveness.

A similar pattern appears when the CrossEncoder is applied to the hybrid candidate pool. Pure CrossEncoder re-ranking over BM25@100 and Dense@1000 obtains MRR@10 = 0.0384, nDCG@10 = 0.0493, and Recall@100 = 0.1867. Again, Recall@100 improves over BM25, but MRR@10 and nDCG@10 decline substantially. These results suggest that the neural model is useful for recall-oriented re-ranking but unstable as a full replacement for the original lexical ranking.

D. Rank-Preserving Fusion

Rank-preserving fusion stabilizes the CrossEncoder by combining the original rank prior with the neural score. The fusion weight λ controls the strength of the original rank signal. Lower values give the CrossEncoder more influence, while higher values preserve BM25 or hybrid ordering more strongly.

The λ sweep shows that the CrossEncoder is useful only when heavily constrained by the original rank prior. With $\lambda=0.9$ the hybrid fusion system obtains MRR@10 = 0.0626 and nDCG@10 = 0.0686, below the BM25 baseline. Increasing λ to 0.95 improves MRR@10 to 0.0700 and nDCG@10 to 0.0755. The best top-rank result is obtained with $\lambda = 0.99$, which gives MRR@10 = 0.0733 and nDCG@10 = 0.0766. This is a small improvement over BM25, but it indicates that the neural model is most effective as a weak corrective signal rather than as a dominant ranking function. Table II shows re-ranking results in detail.

E. Controlled Re-Ranking Variants

Several controlled variants were evaluated to understand why CrossEncoder re-ranking fails. First, a stronger BGE re-ranker was tested with the same rank-preserving fusion setup. It obtains MRR@10 = 0.0708, nDCG@10 = 0.0745, and Recall@100 = 0.1533, which does not improve over the MiniLM CrossEncoder. This suggests that the limitation is not only model size or architecture, but a broader mismatch between off-the-shelf re-rankers and ToT relevance.

Second, rank injection was tested by adding BM25 rank information directly to the CrossEncoder input. This produces MRR@10 = 0.0355, nDCG@10 = 0.0476, and Recall@100 = 0.1867. The result is close to pure CrossEncoder behavior and does not make the neural model reliably BM25-aware.

Third, confidence-gated re-ranking was tested to prevent low-confidence CrossEncoder promotions. A conservative configuration obtains MRR@10 = 0.0541, nDCG@10 = 0.0639, and Recall@100 = 0.1600. A looser configuration obtains MRR@10 = 0.0526, nDCG@10 = 0.0658, and Recall@100 = 0.1533. Both settings remain below the BM25 baseline in top-rank metrics, suggesting that CrossEncoder confidence is not well aligned with ToT relevance.

Fourth, field-aware re-ranking was tested using compact document representations such as title, abstract-like sections, synopsis, and selected metadata. This performs poorly, with MRR@10 = 0.0135, nDCG@10 = 0.0194, and Recall@100

TABLE II
MAIN RETRIEVAL AND RE-RANKING RESULTS ON THE TREC TIP-OF-THE-TONGUE DEVELOPMENT SET.

Method	Candidate Source	Re-ranking Strategy	MRR@10	nDCG@10	R@100
BM25 Baseline	BM25@1000	None	0.0710	0.0747	0.1533
Dense Retrieval	Dense@1000	None	0.0131	0.0178	0.0667
Hybrid Union	BM25@100 $\cup$ Dense@1000	None	0.0710	0.0747	0.1533
Hybrid RRF	BM25 + Dense	RRF	0.0595	0.0706	0.1533
Hybrid + Pure CE	Hybrid@1000	Pure CrossEncoder	0.0384	0.0493	0.1867
BM25@1000 + Pure CE	BM25@1000	Pure CrossEncoder	0.0256	0.0366	0.2133
Hybrid + CE Fusion	Hybrid@1000	$\lambda = 0.99$	0.0733	0.0766	0.1533
Hybrid + Multiview Fusion	Hybrid@1000	Multiview CE, $\lambda = 0.99$	0.0733	0.0766	0.1600

= 0.1333. This indicates that ToT clues may be distributed across longer document text and are not reliably captured by compact structured fields alone.

Finally, multiview CrossEncoder fusion provides the best balanced result. Using multiple document views and max aggregation with $\lambda = 0.99$ gives MRR@10 = 0.0733, nDCG@10 = 0.0766, and Recall@100 = 0.1600. The max aggregation strategy performs better than max mean, which obtains MRR@10 = 0.0700, nDCG@10 = 0.0741, and Recall@100 = 0.1600. This suggests that the relevant ToT clue may appear in a specific document view, and selecting the strongest view-specific signal is more effective than averaging across views.

F. Statistical Significance

The best top-rank system is the multiview CrossEncoder fusion model with $\lambda = 0.99$. Compared with BM25, it improves MRR@10 from 0.0710 to 0.0733 and nDCG@10 from 0.0747 to 0.0766. However, Wilcoxon signed-rank tests show that these gains are not statistically significant. For both MRR@10 and nDCG@10, the p-value is 0.8715, with 2 wins, 147 ties, and 1 loss.

This statistical result clarifies the main conclusion of the experiments. Candidate generation improvements are statistically significant, especially for larger BM25 pools and hybrid candidate union. In contrast, top-rank improvements from neural re-ranking are small and not statistically significant. Therefore, the final finding is not that neural re-ranking strongly outperforms BM25, but that ToT retrieval exposes a trade-off: aggressive neural re-ranking improves Recall@100, while conservative neural fusion protects MRR@10 and nDCG@10 but yields only marginal gains.

VI. DISCUSSION

The results show that the main challenge in ToT retrieval is not a single failure of either BM25 or neural re-ranking. Instead, the retrieval pipeline fails at different stages for different types of queries. Candidate generation determines if the relevant document is available at all while neural re-ranking determines whether the available candidate can be promoted to higher ranks without damaging useful lexical rankings. The difference between CrossEncoder re-ranking and fusion is especially important. Pure neural re-ranking improves recall metric, but fusion produces more stable top-rank behavior.

A. Interpretability Analysis

To investigate the matching behavior of the neural retrieval model, I use contrastive rank-movement analysis. Since the system is a retrieval and re-ranking pipeline, the most suitable explanation is not a single token-level attribution score, but the movement of the known relevant document across retrieval stages. For each query, I compare the rank of the relevant document under BM25, pure CrossEncoder re-ranking, and rank-preserving CrossEncoder fusion. This reveals whether the neural model promotes, demotes, or fails to affect the target document.

The rank-movement analysis displays that the CrossEncoder behaves as a recall-oriented semantic matcher rather than a reliable top-rank replacement for BM25. In some cases, it recognizes relevant documents that BM25 ranks very deeply. For instance, for query 660, the relevant document *Æon Flux* appears at BM25 rank 904, but the CrossEncoder promotes it to rank 50. This explains why pure CrossEncoder re-ranking improves Recall@100. However, the rank-fusion model keeps the same document near its original rank because the lexical prior is very strong. This shows the main trade-off of conservative fusion. It protects the ranking from neural noise but it can also repress useful large promotions.

The opposite pattern also appears. For query 303, BM25 ranks the relevant document *Extracted* at rank 1, but pure CrossEncoder re-ranking demotes it to rank 93. Rank fusion restores it to rank 1. This shows that the CrossEncoder can over-promote semantically plausible distractors even when BM25 has already found the correct document. The aggregate results reflect the same behavior: pure CrossEncoder re-ranking increases Recall@100 but substantially reduces MRR@10 and nDCG@10, while rank fusion improves top-rank stability.

These examples suggest that the CrossEncoder’s matching behavior is not random. It often identifies broad semantic compatibility between the query and candidate documents, which helps when the relevant document is deeply ranked. However, ToT queries are indirect and underspecified, so many incorrect documents can appear semantically plausible. Without a strong lexical prior, the CrossEncoder cannot reliably distinguish the exact known item from these distractors.

B. Error Analysis

The error analysis isolates specific query-document pairs and assigns each query to a failure mode based on rank

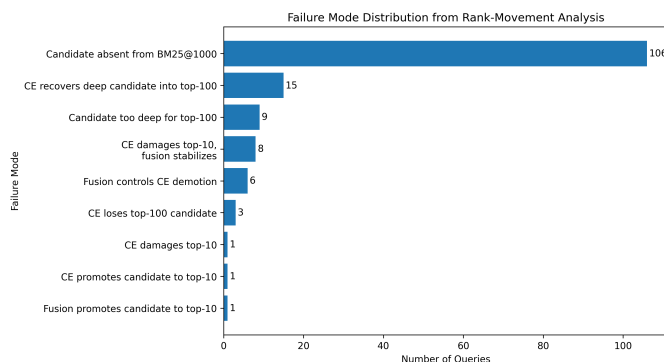


Fig. 1. Failure-mode distribution from contrastive rank-movement analysis.

movement. The largest failure mode is candidate absence. In the local rank-movement analysis, 106 out of 150 queries have no relevant document in the BM25@1000 pool. This confirms that first-stage retrieval is the dominant bottleneck. In most failed cases, the re-ranker cannot help because the target document is not available in its input.

Among the remaining queries, the CrossEncoder exhibits both useful and harmful behavior. It recovers 15 deep candidates into the top 100, showing that neural re-ranking can improve recall when the relevant document is present but deeply ranked. However, it also damages strong BM25 rankings. Eight cases show the pattern where pure CrossEncoder re-ranking damages a top-10 BM25 result, while rank fusion stabilizes the ranking. This supports the main interpretation of the results. The neural model is helpful as a semantic recall signal, but risky as an unconstrained ranker.

Figure 1 summarizes the failure-mode distribution. Candidate absence is by far the most frequent failure mode, followed by CrossEncoder recovery of deep candidates and cases where the relevant document is too deep for the top-100 evaluation cutoff. The smaller categories show neural instability: the CrossEncoder sometimes loses a top-100 BM25 candidate, demotes a top-10 result, or requires rank fusion to control its demotion.

These cases reveal three limitations of the approach. First, candidate expansion is necessary because many target documents are not available to the re-ranker. Second, the CrossEncoder’s semantic matching can recover deeply ranked documents but is not sufficiently calibrated for exact known-item retrieval. Third, rank fusion stabilizes neural re-ranking but also limits the model’s ability to make large beneficial promotions.

C. Implications and Limitations

The main implication is that ToT retrieval requires a different balance between lexical and semantic matching than standard passage re-ranking. In this task, semantic plausibility is not adequate. A document can be related to the query description without being the exact remembered item. This explains why off-the-shelf CrossEncoder re-ranking improves Recall@100 while harming MRR@10 and nDCG@10.

The experiments also show that dense retrieval and neural re-ranking should be used very carefully. Dense retrieval is weak as a standalone ranker but valuable as a complimentary candidate source. Similarly, the CrossEncoder is weak as a full ranking replacement but useful as a correction signal. The best systems therefore preserve the lexical prior instead of replacing it.

A limitation of this project is that the neural re-ranker is not fine-tuned on ToT-specific data. Fine-tuning on the same development set used for evaluation would introduce data leakage while splitting the query set of 150 queries would leave too little data for stable testing. Therefore, this project focuses on off-the-shelf neural re-rankers and diagnostic analysis rather than supervised task adaptation.

VII. CONCLUSION

This project investigated BM25, dense retrieval, hybrid candidate generation, and CrossEncoder-based re-ranking for TREC Tip-of-the-Tongue retrieval. The results show that the main bottleneck is twofold. Many relevant documents are absent from shallow candidate pools, and off-the-shelf neural re-rankers are not reliably calibrated for indirect known-item descriptions. Expanding the BM25 candidate pool and adding dense candidates significantly improves candidate availability, but these gains do not automatically translate into stronger top-rank effectiveness.

Pure CrossEncoder re-ranking improves Recall@100, showing that the neural model can recover some deeply ranked relevant documents. However, it substantially harms MRR@10 and nDCG@10 by over-promoting semantically plausible distractors. Rank-preserving fusion and multiview scoring stabilize this behavior and produce the best top-rank results, but the gains over BM25 remain small and statistically non-significant. Overall, the findings suggest that ToT retrieval requires both stronger candidate generation and task-specific re-ranker calibration. Future work should explore supervised ToT-aware re-ranking with a properly separated training and evaluation setup.

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